

METROPOLITAN UNIVERSITY MEDICAL CENTER

Integrated Electronic Health Record • Clinical Summary Packet

1. Patient Demographics & Administrative Data

FULL NAME Vandelaer, Arthur Michael	PATIENT ID PT-88214-X	DATE OF BIRTH 11/14/1958	AGE / GENDER 67 Years / Male
PRIMARY CARE PHYSICIAN Dr. Evelyn Thorne, MD (Internal Med)	INSURANCE PROVIDER Apex Health Premium PPO	POLICY / GROUP ID AXH-992104-B / GRP-7721	MARITAL STATUS Married
EMERGENCY CONTACT Eleanor Vandelaer (Spouse) • Ph: (555) 019-2834		RESIDENTIAL ADDRESS 1442 Elmcrest Lane, Suite B, Metropolis, NY 10021	

2. Unified Summary Problem List

The following table represents a concise summary of the patient's active, verified clinical conditions as maintained within the core EHR problem list registry.

ICD-10 Code	Condition Description	Onset Date	Status	Documentation MD
E11.9	Type 2 Diabetes Mellitus (without complication)	04-12-2015	Active / Stable	Dr. E. Thorne
I10	Essential (Primary) Hypertension	10/22/2012	Active / Uncontrolled	Dr. E. Thorne
E78.5	Hyperlipidemia, Unspecified	08/14/2016	Active / Controlled	Dr. E. Thorne
N18.31	Chronic Kidney Disease, Stage 3a (Mild-Moderate)	2021-05-19	Active / Chronic	Dr. S. Vance
M10.9	Gout, Unspecified	11/02/2022	Intermittent	Dr. E. Thorne
Z98.89	History of Partial Menisectomy (Left Knee)	03/15/2018	Resolved	Dr. R. Marcus

3. Documented Allergies & Adverse Reactions

Allergen / Agent	Reaction Type	Severity / Clinical Manifestation	Logged Date
Penicillin G / VK	Immunologic (Type I)	Severe anaphylactoid reaction, diffuse urticaria, dyspnea	06/14/1984
Shellfish (Crustaceans)	Hypersensitivity	Moderate GI distress, nausea, localized angioedema (lip swelling)	10/05/1999
Seasonal Allergen (Tree Pollen)	Environmental	Mild allergic rhinitis, conjunctival injection, sneezing	04/12/2010

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4. Outpatient Medication Profile

The following profile combines verified current prescriptions, active home medications, and recent modifications. *AI System Evaluation Prompt: Review carefully for therapeutic duplications and potential drug-drug or drug-disease interaction profiles.*

Medication Name	Dosage / Strength	Frequency / Route	Prescribing Provider	Indication / Notes
Metformin HCl	500 mg	Oral, twice daily with meals	Dr. Evelyn Thorne, MD	Type 2 Diabetes Mellitus
Lisinopril	20 mg	Oral, once daily (morning)	Dr. Evelyn Thorne, MD	Hypertension / Renal Protection
Atorvastatin Calcium	40 mg	Oral, once daily at bedtime	Dr. Evelyn Thorne, MD	Hyperlipidemia / ASCVD Risk
Amlodipine Besylate	5 mg	Oral, once daily	Dr. Evelyn Thorne, MD	Hypertension Adjunct Therapy
Furosemide	20 mg	Oral, once daily (morning)	Dr. Evelyn Thorne, MD	Fluid management / Hypertension
Metformin	500 mg	Take 1 tablet twice a day	Dr. Evelyn Thorne, MD	[DUPLICATE ENTRY IN EHR]
Allopurinol	100 mg	Oral, once daily	Dr. Evelyn Thorne, MD	Gout Prophylaxis
Ibuprofen	400 mg	Oral, every 6 hours as needed for pain	Dr. Evelyn Thorne, MD	[MILD RISK] Left knee osteoarthritic pain
Sitagliptin Phosphate	100 mg	Oral, once daily	Dr. Evelyn Thorne, MD	Type 2 Diabetes Mellitus
Cyanocobalamin (Vit B12)	1000 mcg	Oral, once daily	Dr. Evelyn Thorne, MD	Nutritional Supplementation
Cholecalciferol (Vit D3)	2000 IU	Oral, once daily	Dr. Evelyn Thorne, MD	Vitamin D Insufficiency

Clinical Pharmacist Reconciliation Note: Patient states he fills all prescriptions at Community Care Pharmacy. A potential conflict exists with the concurrent prescription of *Ibuprofen 400mg PRN* in the setting of *Chronic Kidney Disease Stage 3a* and concurrent *Lisinopril* therapy. Monitor renal function closely. Furthermore, Metformin is listed twice via distinct entry streams, indicating a systemic reconciliation artifact.

5. Extended Medical History & Hospitalizations

Chronic Disease Trajectory

Type 2 Diabetes Mellitus: Diagnosed in 2015 during routine screening. Managed initially via lifestyle modification and Metformin monotherapy. Sitagliptin added in 2023 due to ascending HbA1c values. Complies moderately with dietary restrictions, though reports occasional carbohydrate overindulgence.

Cardiovascular & Renal Profile: Longstanding Stage 2 hypertension. Blood pressure control has been historically volatile, requiring multi-drug regimens. Developed secondary hyperlipidemia, managed via moderate-to-high intensity statin therapy. Baseline serum creatinine has shown progressive elevation over the past 48 months, establishing stable Chronic Kidney Disease (CKD) Stage 3a.

Past Surgical & Hospitalization History

- **Hospitalization (October 14, 2024):** Admitted to Metropolitan Emergency Department following acute onset of severe right first metatarsophalangeal (MTP) joint pain and swelling. Diagnosed with acute gouty arthropathy flair. Resolved via short-course colchicine and temporary corticosteroid therapy.
- **Outpatient Surgical Intervention (March 15, 2018):** Left knee arthroscopy with partial medial menisectomy following structural tear during recreational activity. Performed by Dr. R. Marcus at Peak Orthopedics. Uncomplicated recovery.

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6. Recent Laboratory Diagnostics

Specimen Collection Date: May 12, 2026 | **Status:** Final Verified | **Lab Facility:** MUMC Central Pathology Group

Complete Blood Count (CBC)

Component Lab Test	Result Value	Reference Range	Status Flags
White Blood Cell (WBC)	6.8 x10 ³ / µL	4.5 - 11.0 x10 ³ / µL	Normal
Red Blood Cell (RBC)	4.42 x10 ⁶ / µL	4.30 - 5.90 x10 ⁶ / µL	Normal
Hemoglobin (Hgb)	13.8 g/dL	13.5 - 17.5 g/dL	Normal
Hematocrit (Hct)	41.2 %	41.0 - 50.0 %	Normal
Platelets	245 x10 ³ / µL	150 - 450 x10 ³ / µL	Normal

Comprehensive Metabolic & Lipid Panels

Component Lab Test	Result Value	Reference Range	Status Flags
Fast Serum Glucose	146 mg/dL	70 - 99 mg/dL	HIGH (Abnormal)
Hemoglobin A1c (HbA1c)	7.9 %	4.0 - 5.6 %	HIGH (Abnormal)
Serum Creatinine	1.52 mg/dL	0.70 - 1.30 mg/dL	HIGH (Abnormal)
eGFR (CKD-EPI)	44 mL/min/1.73m ²	>= 60 mL/min/1.73m ²	LOW (Abnormal)
Serum Sodium	139 mEq/L	136 - 145 mEq/L	Normal
Serum Potassium	4.6 mEq/L	3.5 - 5.1 mEq/L	Normal
Total Cholesterol	178 mg/dL	< 200 mg/dL	Normal
LDL Cholesterol (Calculated)	98 mg/dL	< 100 mg/dL	Normal (Targeted)
HDL Cholesterol	42 mg/dL	> 40 mg/dL	Normal
Triglycerides	190 mg/dL	< 150 mg/dL	ELEVATED

Ambiguous Reference Lab Note: An isolated testing stream labeled simply "GLU-M-2" was executed on a separate sample subset on **05/14/2026**, indicating a value of **112** without specifying standard units (mg/dL vs mmol/L) or fasting parameters, though annotated "post-prandial evaluation requested by outside clinic." AI systems parsing this should flag the baseline clinical ambiguity.

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7. Longitudinal Progress Notes

The following chronic care and targeted follow-up encounter notes are retrieved sequentially from the primary care and specialized services modules.

Encounter Date: 02/10/2026 • Primary Care Follow-Up • Provider: Evelyn Thorne, MD

Chief Complaint: Routine 3-month management follow-up for multiple chronic medical conditions, including type 2 diabetes mellitus, Stage 2 hypertension, and chronic kidney disease.

Subjective: Patient reports general well-being but admits to intermittent compliance with low-sodium and low-carbohydrate nutritional recommendations. Describes mild bilateral lower extremity fatigue after walking prolonged distances. No chest pain, orthopnea, or acute dyspnea. Home blood pressures ranges from 142/88 to 155/92 mmHg according to self-logged logs.

Objective Evaluation:

- Vitals: BP 148/90 mmHg, HR 72 bpm (regular), Temp 98.4°F, Wt 198 lbs, SpO2 97% on room air.
- Heart: Regular rhythm, normal S1/S2. No murmurs or gallops observed.
- Lungs: Clear to auscultation bilaterally; no wheezes, rales, or rhonchi.
- Extremities: 1+ trace pitting edema over bilateral pretibial regions. Distal pulses palpable.

Assessment & Plan:

1. *Hypertension:* Suboptimally controlled. Decided to increase Amlodipine dosage from 5 mg to 10 mg daily to achieve a target BP < 130/80 mmHg given renal status. Continued Lisinopril 20 mg.
2. *Type 2 Diabetes:* Glycemic control suboptimal. Patient's historical home fingerstick logs reflect elevated morning readings (140-160 mg/dL). Will maintain current Metformin and Sitagliptin pending comprehensive lab updates next quarter.

Encounter Date: 04-03-2026 • Telehealth Telemedicine Urgent Visit • Provider: Marcus Sterling, NP

Chief Complaint: Pronounced fatigue, muscle weakness, and polyuria over the past ten days.

Subjective: Patient requested an urgent telehealth visit complaining of persistent lethargy that is interfering with his standard daily walks. States he has noticed an increasing frequency of urination, particularly nocturnal episodes (3-4 times per night). Denies dysuria, hematuria, fevers, or chills. Wondered if symptoms were related to a recent upper respiratory exposure.

Objective Evaluation: Visual inspection via secure high-definition video link shows an alert, oriented male in no acute respiratory distress. Appears mildly fatigued. Moist mucous membranes. No obvious facial asymmetry or respiratory accessory muscle utilization.

Assessment & Plan:

1. *Fatigue & Polyuria Protocol:* Symptoms strongly point toward escalating hyperglycemia vs. electrolyte imbalance. Ordered immediate comprehensive lab workup including fasting plasma glucose, basic metabolic panel, and HbA1c.

2. *EHR Inconsistency/Discrepancy Note:* **[CONFLICTING NOTE]** Based on visual assessment and standard protocol, patient's blood pressure is estimated as stable; medication alterations are deferred. (*Note: Electronic chart indicates BP was deferred due to virtual nature, yet NP notes "stable vascular control maintained on present regimen", directly contradicting the subsequent clinic visit findings of profound hypertension*). Ordered urgent in-clinic follow-up with PCP.

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Encounter Date: 2026-05-15 • Post-Lab Comprehensive Review • Provider: Evelyn Thorne, MD

Chief Complaint: Detailed review of recent laboratory panels and intensive care coordination for uncontrolled hyperglycemia and elevated renal markers.

Subjective: Patient returns following completion of labs on May 12, 2026. Acknowledges feeling "run down" and chronically tired. Discussed home medication compliance; patient states he takes all medications regularly but notes he has been taking OTC Ibuprofen 400 mg multiple times a week to manage chronic left knee osteoarthritic discomfort.

Objective Evaluation:

- Vitals: BP 154/92 mmHg (Confirmed high), HR 76 bpm, Wt 199 lbs.
- Critical Labs Reviewed: HbA1c 7.9%, Fasting Glucose 146 mg/dL, Serum Creatinine 1.52 mg/dL, eGFR 44 mL/min/1.73m².

Clinical Synthesis & Plan:

1. *Renal Impairment & Medication Safety:* The combination of elevated creatinine and lowered eGFR represents a progression of CKD Stage 3a. Advised patient to immediately cease all over-the-counter NSAID (Ibuprofen) consumption due to substantial risk of worsening nephrotoxicity, especially when coupled with his ACE inhibitor (Lisinopril). Will refer to Orthopedics for non-pharmacologic or alternative analgesic management of left knee osteoarthritic pain.
2. *Hyperglycemia:* HbA1c is poorly controlled at 7.9%. Counseled extensively on carbohydrate restrictions. Will continue Metformin 500mg BID and Sitagliptin 100mg daily for now, with a plan to re-evaluate for potential SGLT2 inhibitor addition if renal status stabilizes after NSAID cessation.

8. Specialized Imaging & Diagnostic Reports

DOCUMENT TYPE: Diagnostic Ultrasound Report (Abdomen)
EXAMINATION DATE: March 22, 2026
ACCESSION NUMBER: RAD-2026-8871A
ORDERING PHYSICIAN: Dr. Evelyn Thorne, MD
INTERPRETATION BY: Dr. Sarah Vance, MD (Department of Radiology)

Clinical Indication

Persistent right upper quadrant abdominal discomfort, mild nausea, and a history of metabolic syndrome with hyperlipidemia. Evaluate for cholelithiasis vs. hepatic steatosis.

Imaging Technique

Real-time gray-scale and color Doppler transabdominal ultrasound imaging was performed of the upper abdomen according to standard institutional protocols. Comparisons made to prior abdominal imaging dated June 11, 2022.

Findings & Organ Evaluation

Liver: The liver is diffusely increased in echogenicity with mild attenuation of the deeper parenchymal structures and poor visualization of the intrahepatic vascular borders. The liver is smooth-contoured and measured at the upper limits of normal length (16.2 cm), suggestive of mild hepatomegaly. No focal solid or cystic masses are detected within either lobe. Flow within the main portal vein is hepatopetal and demonstrates appropriate velocity.

Gallbladder & Biliary Tree: The gallbladder is well-distended. No shadowing gallstones or sludge layers are visualized. The gallbladder wall is normal in thickness, measuring 2.4 mm, with no focal hyperemia. The sonographic Murphy's sign was negative at the time of evaluation. The common bile duct is normal in caliber, measuring 4.1 mm at the porta hepatis.

Pancreas & Spleen: The visualized portions of the pancreatic head and body appear unremarkable, though the pancreatic tail is obscured by overlying bowel gas. The spleen is normal in size and echotexture, measuring 10.5 cm in bipolar length.

Kidneys: The right kidney measures 9.8 cm and the left kidney measures 10.1 cm. There is mild, symmetric thinning of the renal cortex bilaterally, consistent with the patient's documented clinical history of moderate nephrosclerosis / chronic kidney disease. No hydronephrosis, large obstructing calculi, or suspicious solid renal masses are identified bilaterally.

Diagnostic Impression

1. **Diffuse Hepatic Steatosis:** Correlating with a pattern of moderate fatty liver infiltration. No focal hepatic masses.
2. **No Sonographic Evidence of Acute Cholecystitis or Cholelithiasis:** Completely normal biliary tree and gallbladder morphology.
3. **Bilateral Renal Cortical Thinning:** Nonspecific finding, highly consistent with intrinsic parenchymal medical renal disease.

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9. Inpatient Hospital Discharge Summary

Historical Context Record: This summary details the patient's last major acute care institutional hospitalization prior to the current year's clinic follow-ups. Included to provide historical continuity for AI clinical model processing.

Admission Date: October 14, 2024 | **Discharge Date:** October 17, 2024

Attending Physician: Dr. David Aris, MD (Inpatient Medicine Group)

Principal Discharge Diagnosis: Acute Gouty Arthropathy (Right First MTP Joint Flair)

Secondary Diagnoses: Type 2 Diabetes Mellitus, Essential Hypertension, Chronic Kidney Disease Stage 3a.

Clinical Presentation & Hospital Course

The patient, a 65-year-old male at the time of admission, presented to the Emergency Department with emergency-level acute, agonizing pain, localized erythema, and marked swelling of the right first metatarsophalangeal (MTP) joint. The symptoms began abruptly the previous evening following a family gathering. He was unable to tolerate weight-bearing or even the pressure of a bedsheet. Initial vital signs showed a blood pressure of 162/95 mmHg, heart rate 88 bpm, and temperature 99.1°F.

Laboratory evaluation at admission revealed an elevated serum uric acid level of 9.2 mg/dL and an acute rise in serum creatinine to 1.65 mg/dL above his known baseline. Joint aspiration of the right first MTP was deferred given classic presentation and patient preference; however, clinical criteria firmly established acute podagra. Due to the patient's baseline Chronic Kidney Disease Stage 3a, traditional high-dose nonsteroidal anti-inflammatory drugs (NSAIDs) like Indomethacin were strictly contraindicated.

The patient was admitted to the medicine floor and initiated on an optimized low-dose Colchicine regimen (1.2 mg loading dose followed by 0.6 mg one hour later, then 0.6 mg daily), alongside a short course of oral Prednisone (30 mg daily, tapered over 5 days) to manage the severe localized inflammatory cascade. His hypertension was managed by continuing his home regimen of Lisinopril and Metformin. Over the 72-hour hospital course, the patient demonstrated significant clinical improvement, with near-total resolution of joint pain and full restoration of baseline ambulation.

Discharge Status & Condition

Ambulatory, stable, and in good spirits. Right first MTP joint displays minimal residual erythema and no functional pain. Renal function parameters stable at discharge (Creatinine 1.48 mg/dL).

Discharge Medications & Outpatient Plan

The patient was instructed to resume his chronic home medications exactly as prescribed prior to admission, with the explicit addition of targeted short-term therapies:

- Prednisone 30mg Tablet:** Take 1 tablet daily by mouth for 2 remaining days to complete the prescribed inflammatory post-discharge taper. Discontinue thereafter.
- Colchicine 0.6mg Tablet:** Take 1 tablet daily by mouth for a total of 5 additional days for post-flare suppression, then discontinue.

3. **Allopurinol 100mg Tablet:** Confirmed to resume for long-term uric acid suppression and gout prophylaxis.

Crucial Note: Patient was warned never to escalate Allopurinol dosage during an acute inflammatory flare.

Mandated Follow-Up Instructions

Patient is scheduled to see his Primary Care Physician, Dr. Evelyn Thorne, within 10 to 14 days of discharge for a formal comprehensive metabolic check, repeatable serum creatinine monitoring, and long-term uric acid assessment. Patient was thoroughly counseled on adherence to a low-purine diet, avoidance of high-fructose corn syrup, alcohol moderation, and the critical importance of avoiding over-the-counter NSAIDs due to his delicate renal threshold.

*** END OF DIGITAL HEALTH RECORD PACKET - METROPOLITAN UNIVERSITY MEDICAL CENTER ***

The data contained herein is entirely synthetic and procedurally generated for testing, engineering validation, and offline RAG benchmarking purposes.